THE DATAGOOD(R) PACKAGE: AN OPEN SCIENCE APPROACH TO DATA ENGINEERING

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CUSTOMIZING LAYOUTS

Flexible Layouts

- The RG and DD QMD templates include a built-in style sheet that divides the page into sections
- Each section is populated by variable attributes (text descriptors for dictionary elements or data summaries)
- Each data type leverages different sections of the layout to organize content
- Positions of content and transformer functions are defined in the layout object
- The layout object will be embedded within the QMD template file so that users can adapt as needed
- Custom formatting functions can be defined by the user

```
layout <-
```

```
c( "div2 ;; vlabel  ;; LABEL         ;; v_to_txt",
   "div3 ;; vtype   ;; DATA TYPE     ;; v_to_txt",
   "div3 ;; scope   ;; SCOPE         ;; v_to_txt",
   "div4 ;; desc    ;; DESCRIPTION   ;; v_to_txt",
   "div4 ;; flevels ;; LEVELS        ;; f_to_txt",
   "div4 ;; glevels ;; ''            ;; v_to_txt",
   "div4 ;; loc     ;; LOCATION CODE ;; v_to_txt",
   "div5 ;; v       ;; STATS         ;; get_properties"  )
```

```
|DIV  |VARIABLE  |LABEL         |FORMATTING FUNCTION  |
|:----|:---------|:-------------|:--------------------|
|div2 |vlabel    |LABEL         |v_to_txt             |
|div3 |vtype     |DATA TYPE     |v_to_txt             |
|div3 |scope     |SCOPE         |v_to_txt             |
|div4 |desc      |DESCRIPTION   |v_to_txt             |
|div4 |flevels   |LEVELS        |f_to_txt             |
|div4 |glevels   |''            |v_to_txt             |
|div4 |loc       |LOCATION CODE |v_to_txt             |
|div5 |v         |LABEL         |get_properties       |
```

CSS Elements Controlling Layouts

```
.parent {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  grid-template-rows: auto;
    grid-column-gap: 20px;
  grid-row-gap: 10px;
  grid-template-areas:
    "a a a"
    "b b b"
    "c d d"
    "e f g"
    "h h h"
    "i i i"
    "j k k"
    "m o o"
    "n o o"
    "p p p";
}
```

```
.div1 {
  grid-area: a;
  padding-bottom: 15px;
}

.div2 {
    grid-area: b;
    padding-left: 30px;
}

.div3 {
    grid-area: c;
    padding-left: 30px;
}
```

Data Dictionary Format

- Accelerate the documentation and sharing of a dataset.
- The DGF provides a quick list-style template for providing the most important features of a dataset:
  - variable name
  - label
  - description / details
  - data types
  - factor level definitions
  - custom attributes (e.g. SCOPE)
- The data dictionary (DD) version limits documentation to dataset and variable definitions
- The research guide (RG) version includes data profiles

VarName_01_X

Ipsum is simply dummy text of the printing and typesetting industry.

DATA TYPE: numeric
SCOPE: PZ
LENGTH: 6

DEFINITION: Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book.

VarName_02_L2

Ipsum is simply dummy text of the printing and typesetting industry.

DATA TYPE: factor
SCOPE: PC
LENGTH: 24

DEFINITION: Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book.

LEVELS:

A	goat
B	monkey
C	turtle
D	horse

data dictionary format

VarName_01_X

Ipsum is simply dummy text of the printing and typesetting industry.

DATA TYPE: numeric
SCOPE: PZ
LENGTH: 6

DEFINITION: Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book.

VarName_02_L2

div1

div2 Ipsum is simply dummy text of the printing and typesetting industry.

DATA TYPE: factor
SCOPE: PC
LENGTH: 24

div3

DEFINITION: Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book.

LEVELS:

A	goat
B	monkey
C	turtle
D	horse

div4

data dictionary format

`create_dd( dgf )`

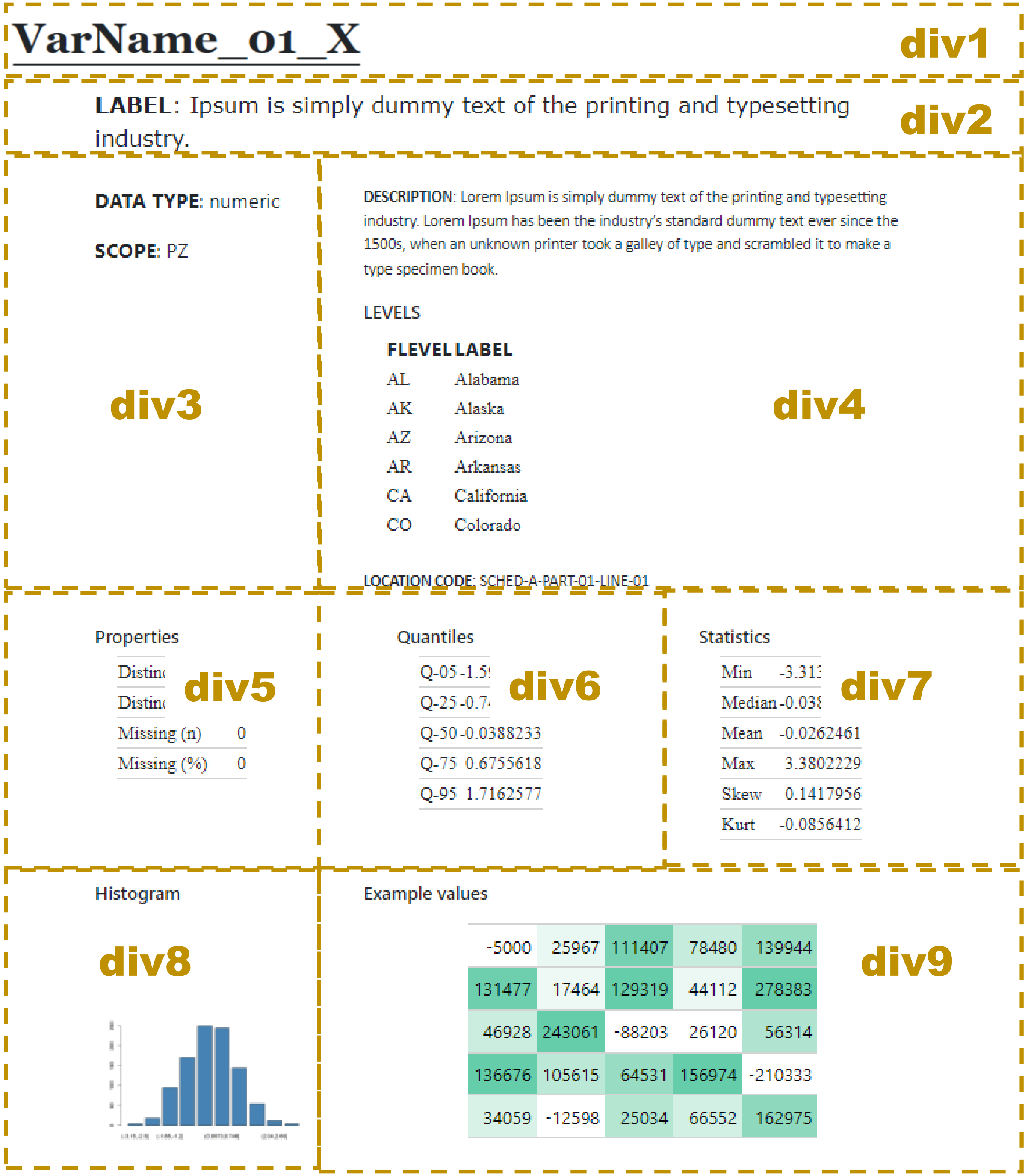
+

`layout`

Research Guide Format

- The research guide provides more thorough documentation of a dataset than dictionary files
- The package manages data profiling functionality
- The QMD document can be turned into a true data user guide or public use file codebook that documents the data generation process and transformation
    - sampling framework, response rates, weights
    - survey questions or raw data fields
    - cleaning, standardization, normalization, and transformation processes applied
    - privacy protecting steps
    - etc

data profiling layouts in the RG.qmd file



data profiling

VarName_01_X

div1

div2

div3

div4

div8

data quality

div9

preview

div10

stats

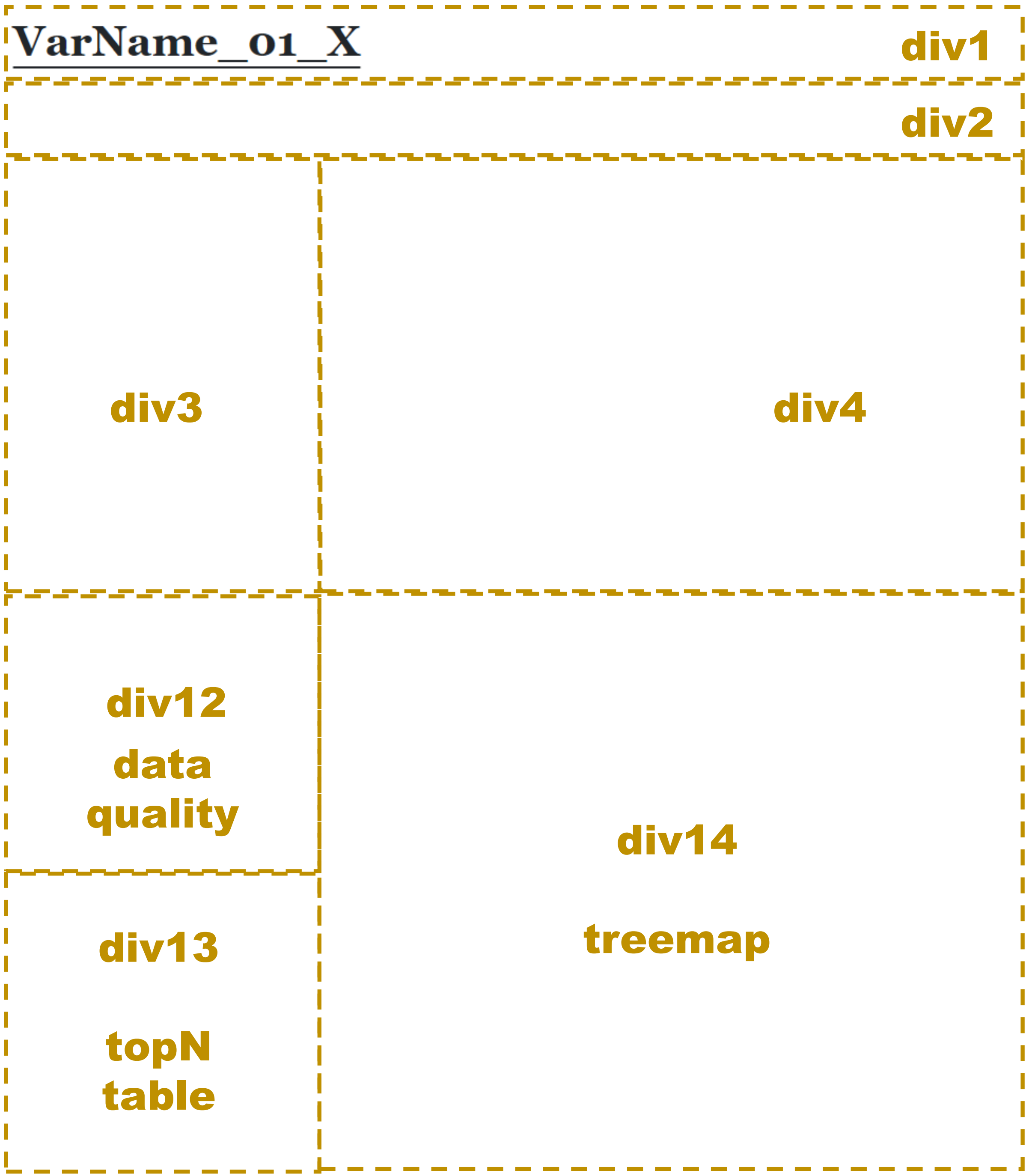
min, ave, max string length

div11

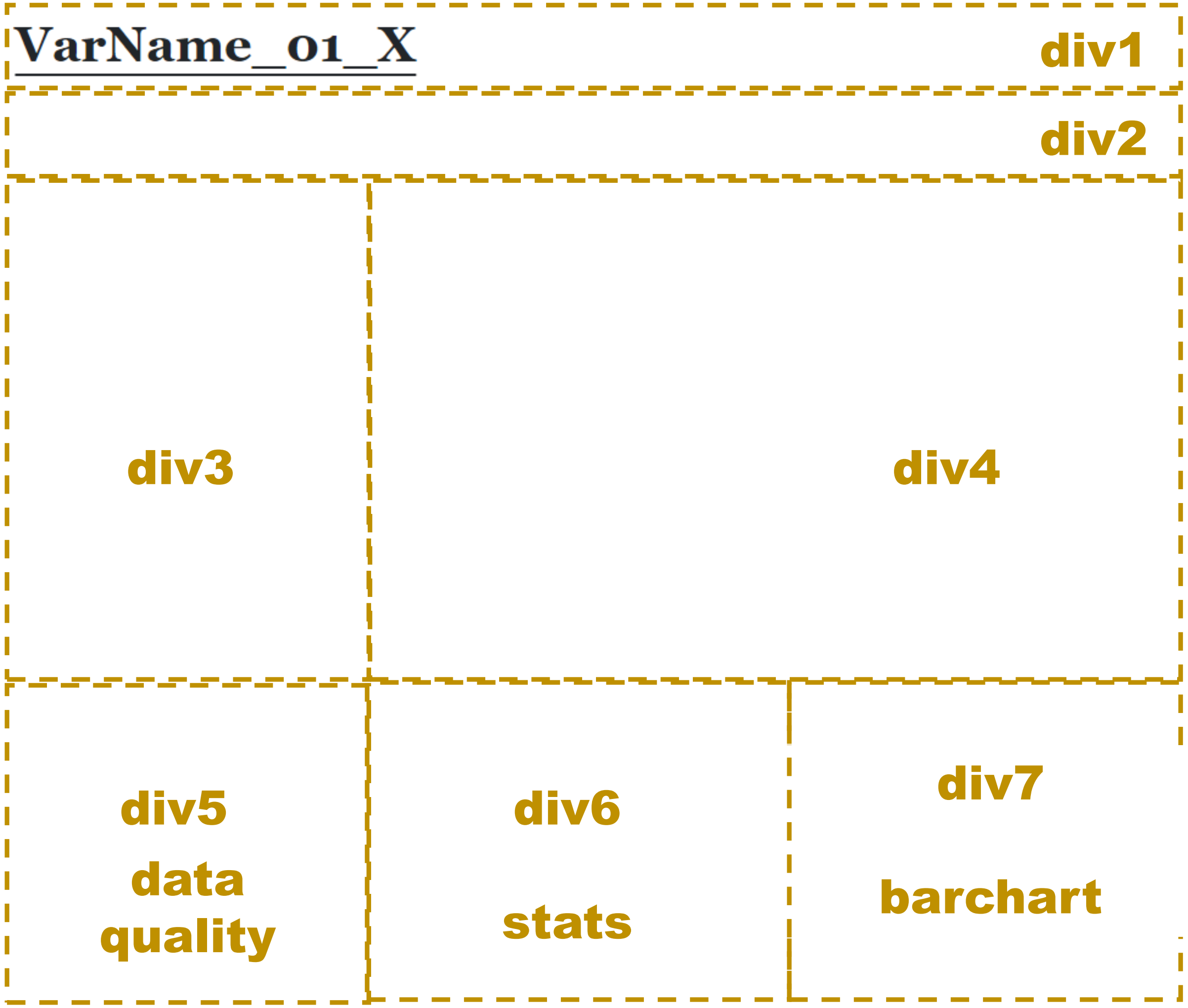
word cloud

data type:

character

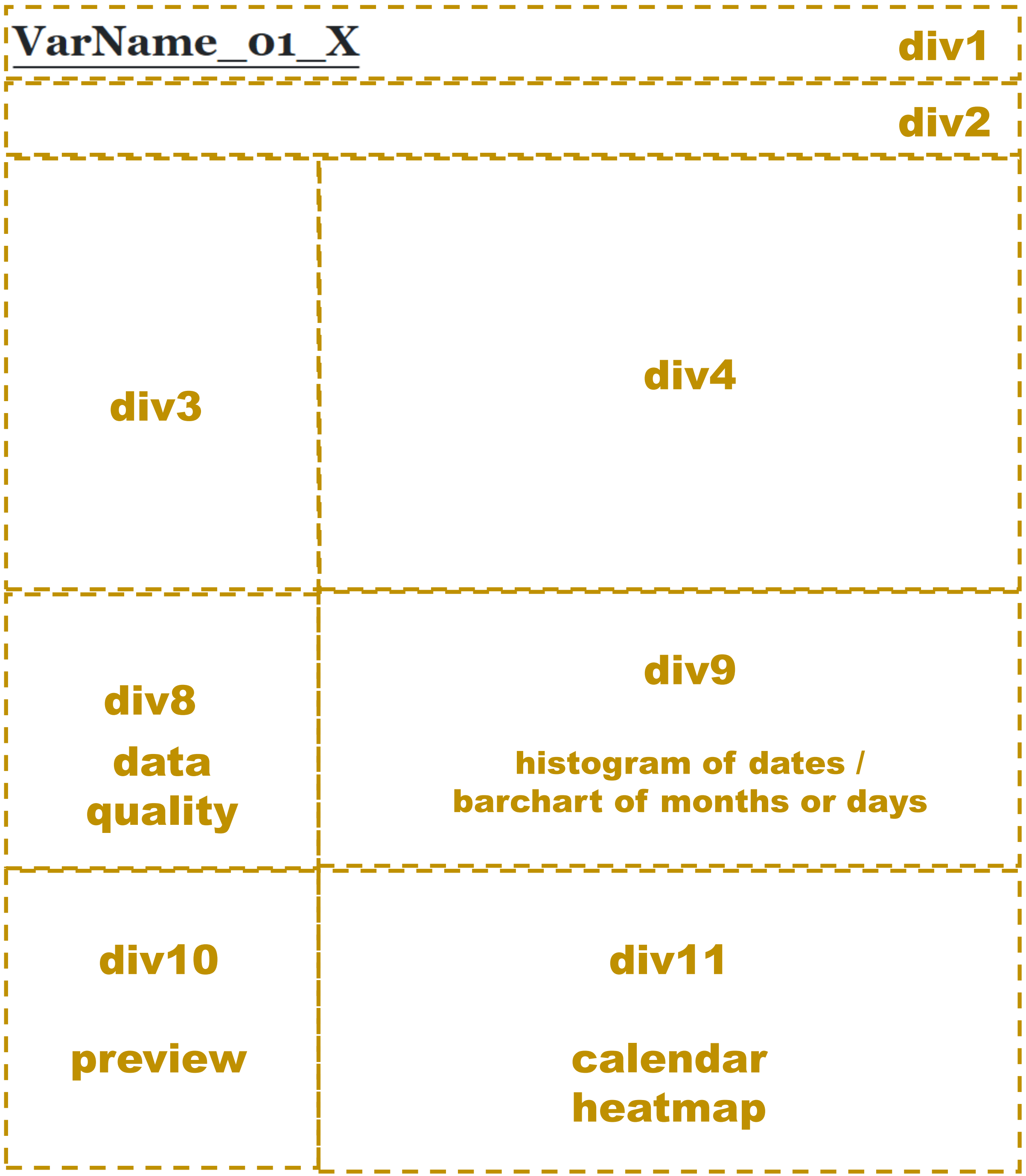


data type:
factor



data type:

logical (Boolean)



data type:

dates / time

ITERATIVE WORKFLOW

create_dgf(df.raw) — Create the DGF (user edits, adds labels and rules)

inspect_dgf(dgf) — Ensure json ok, rules are defined, etc.

ingest_raw(df.raw, dgf) — Apply the raw_convert rules, keep desired types.

data is
altered

update_dgf() Call after changes are made to data to ensure stored values are correct (e.g factor levels the same, update stored descriptive stats for numeric vars).

standardize(df.v2, dgf) — Apply standardization rules, update_dgf() called silently

data is
altered

create_vr(df.v3, dgf) — Create a data validation report using dgf_validate rules.

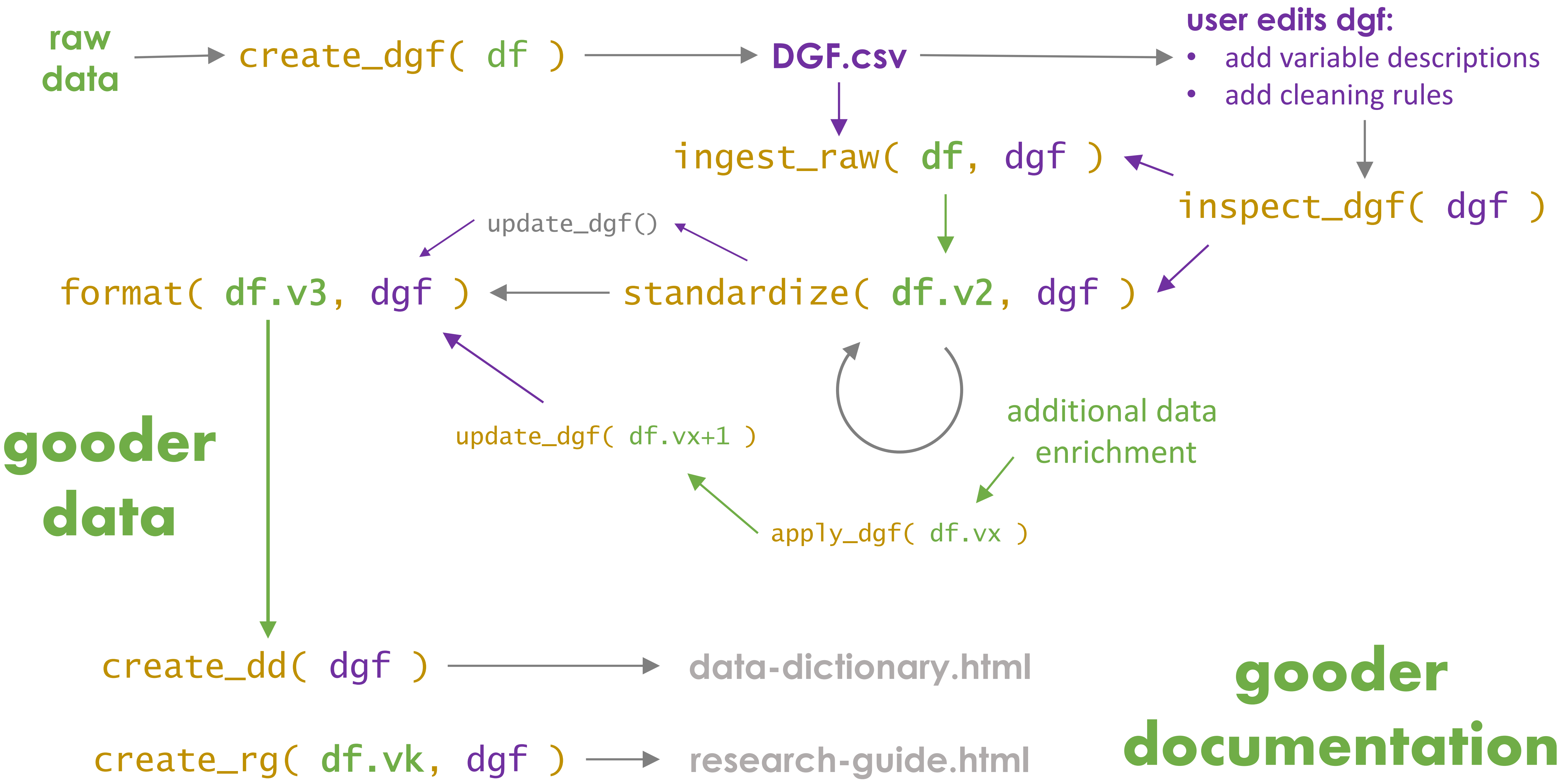
create_dd(df.v3, dgf) — Create a data dictionary from dgf fields.

create_rg(df.v3, dgf) — Create a research guide qmd, annotated before rendering.

can be gener-ated any time once DGF exists

format(df.v3, dgf) — Apply formatting rules to make data pretty.

df.share — some context where the data is read, not analyzed, also used in dd & rg


raw data
create_dgf(df)
DGF.csv
user edits dgf:
• add variable descriptions
• add cleaning rules
ingest_raw(df, dgf)
inspect_dgf(dgf)
update_dgf()
format(df.v3, dgf)
standardize(df.v2, dgf)
update_dgf(df.vx+1)
additional data enrichment
apply_dgf(df.vx)
gooder data
create_dd(dgf)
data-dictionary.html
create_rg(df.vk, dgf)
research-guide.html
gooder documentation

Default Graphical Summaries

Customized Graphics

Flexible Layouts

- Update the layout object by adding or substituting a section (div) label and function name
- Make the function available to the QMD template (not yet implemented)
  - Package will automatically try to source a DG.R script when rendering files. It will contain some of the default formatting and graphing functions for the user to customize, and they can add their own
  - Functions can be added directly to the QMD document and managed there

layout <-

```
c( "div2 ;; vlabel  ;; LABEL         ;; v_to_txt",
   "div3 ;; vtype   ;; DATA TYPE     ;; v_to_txt",
   "div3 ;; scope   ;; SCOPE         ;; v_to_txt",
   "div4 ;; desc    ;; DESCRIPTION   ;; v_to_txt",
   "div4 ;; flevels ;; LEVELS        ;; f_to_txt",
   "div4 ;; glevels ;; ''            ;; v_to_txt",
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|div3 |scope     |SCOPE         |v_to_txt             |
|div4 |desc      |DESCRIPTION   |v_to_txt             |
|div4 |flevels   |LEVELS        |f_to_txt             |
|div4 |glevels   |''            |v_to_txt             |
|div4 |loc       |LOCATION CODE |v_to_txt             |
|div5 |v         |LABEL         |get_properties       |
```
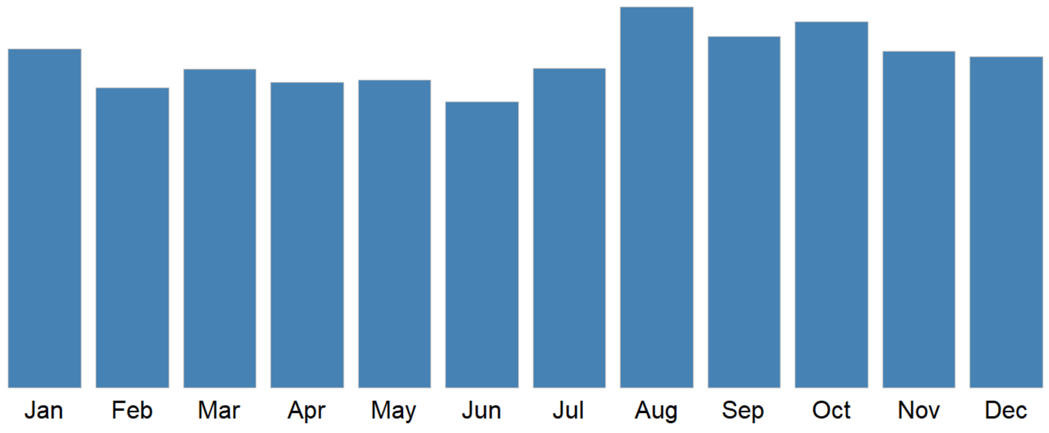
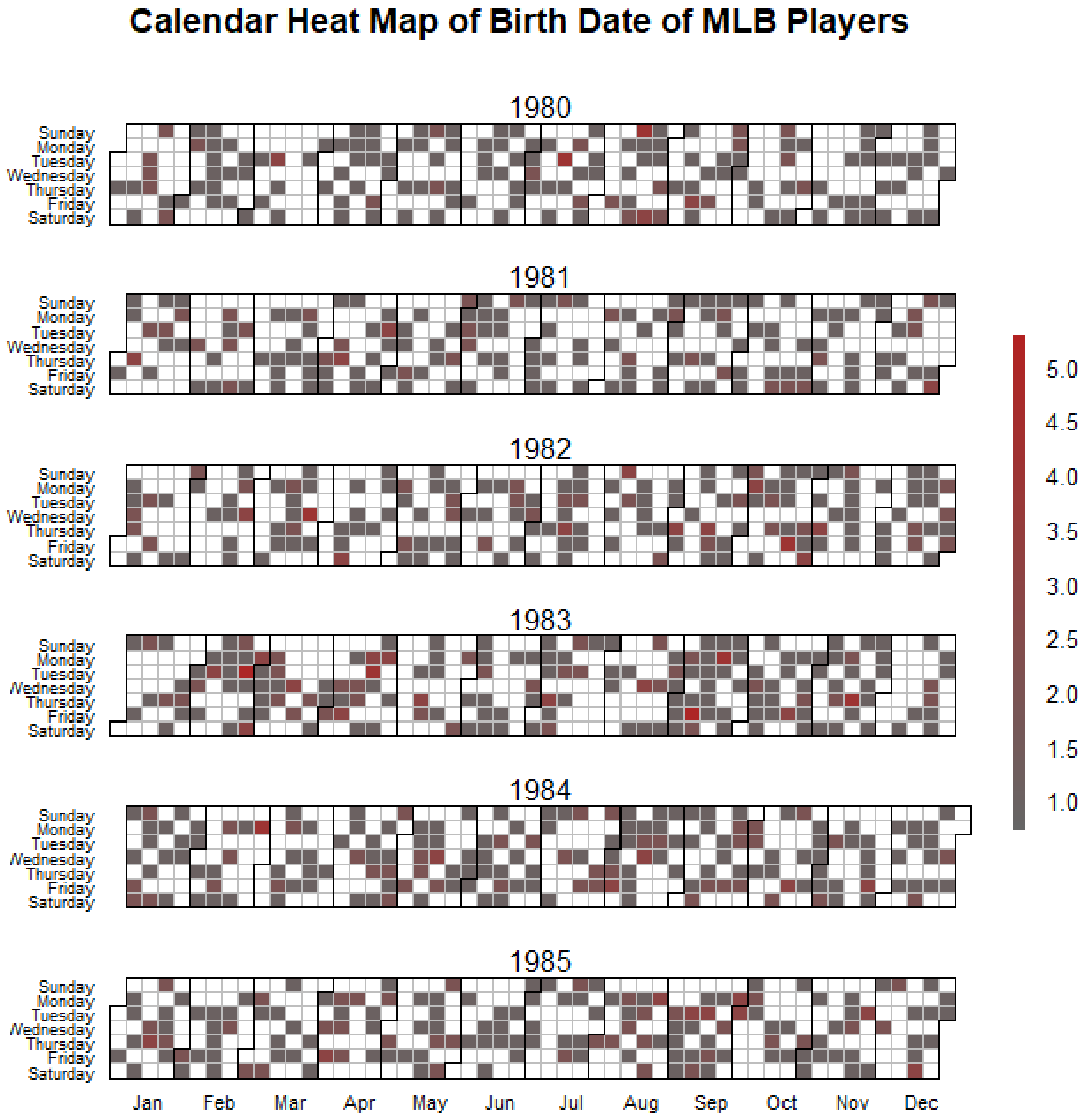
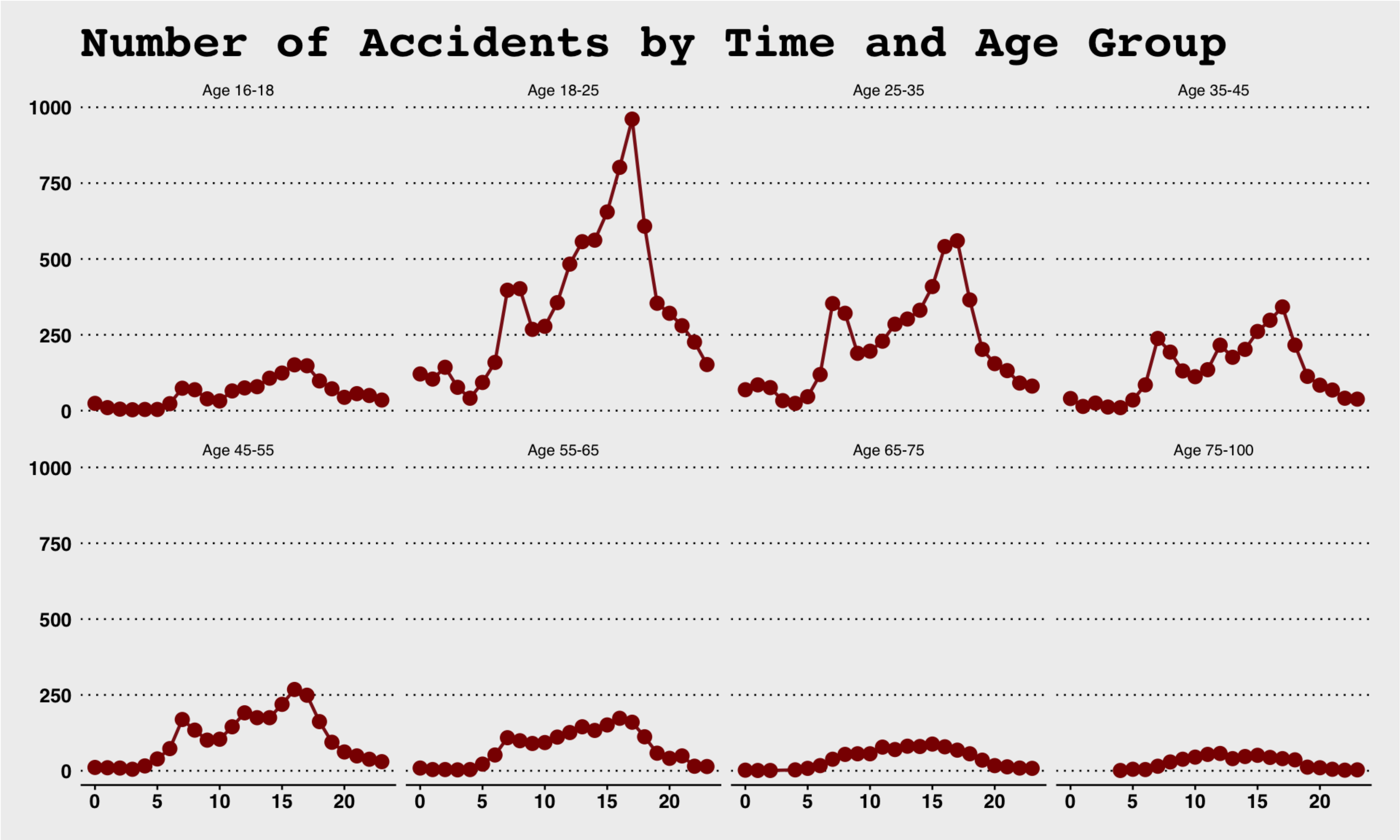

Visualizations for date might be contingent upon the type. Heatmap makes sense for time.



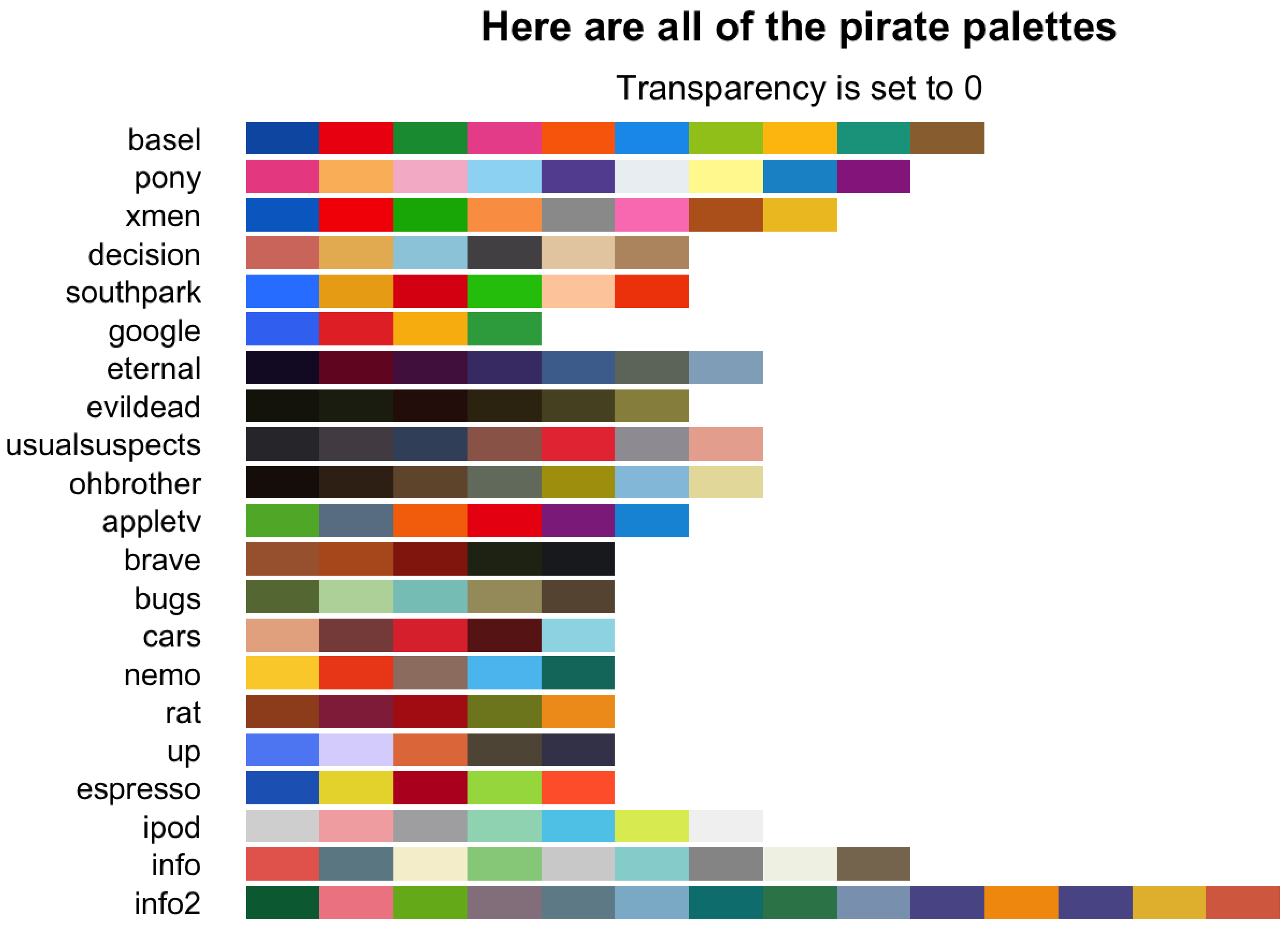


Days of week and hours of the day

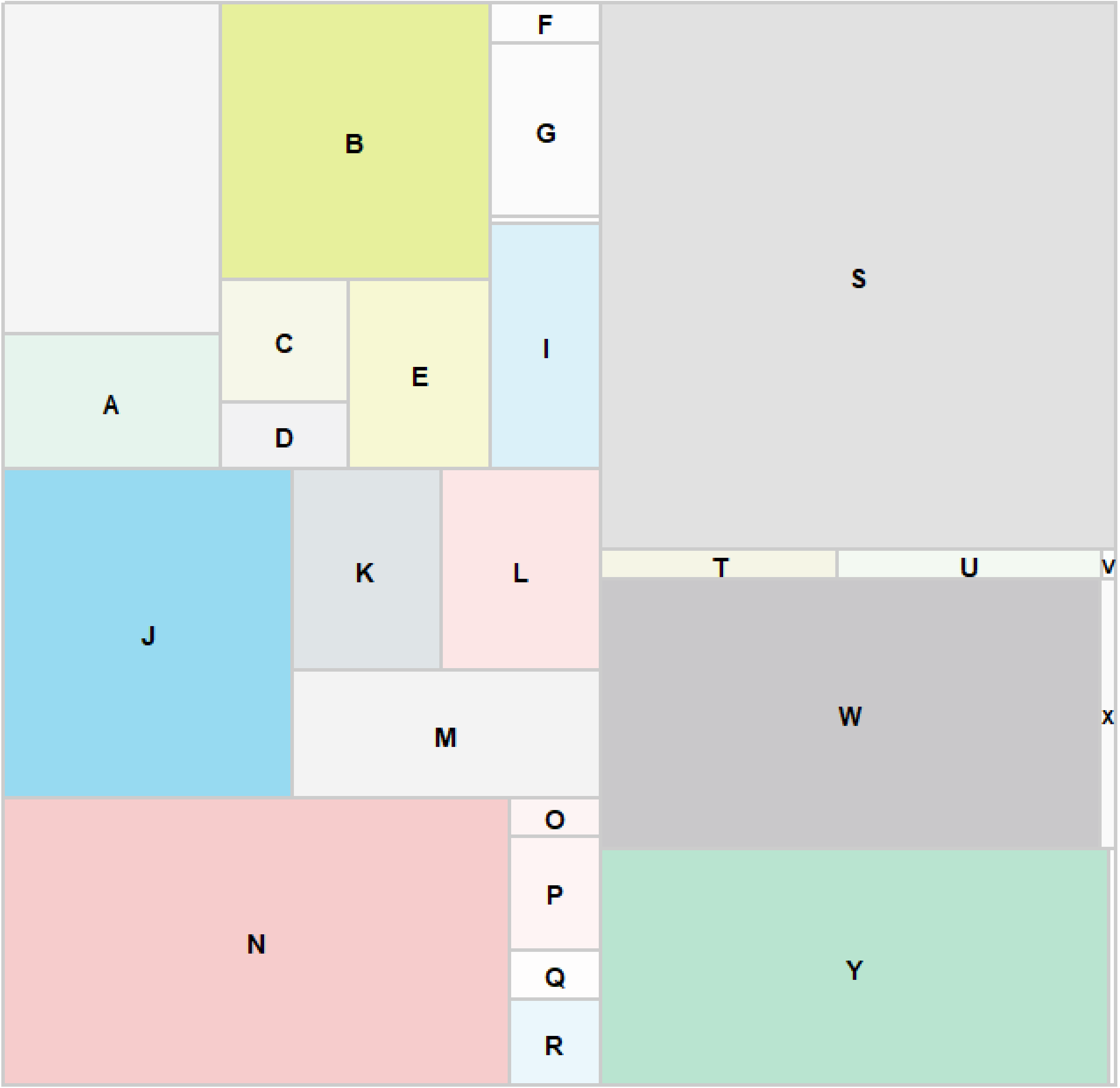


thank you

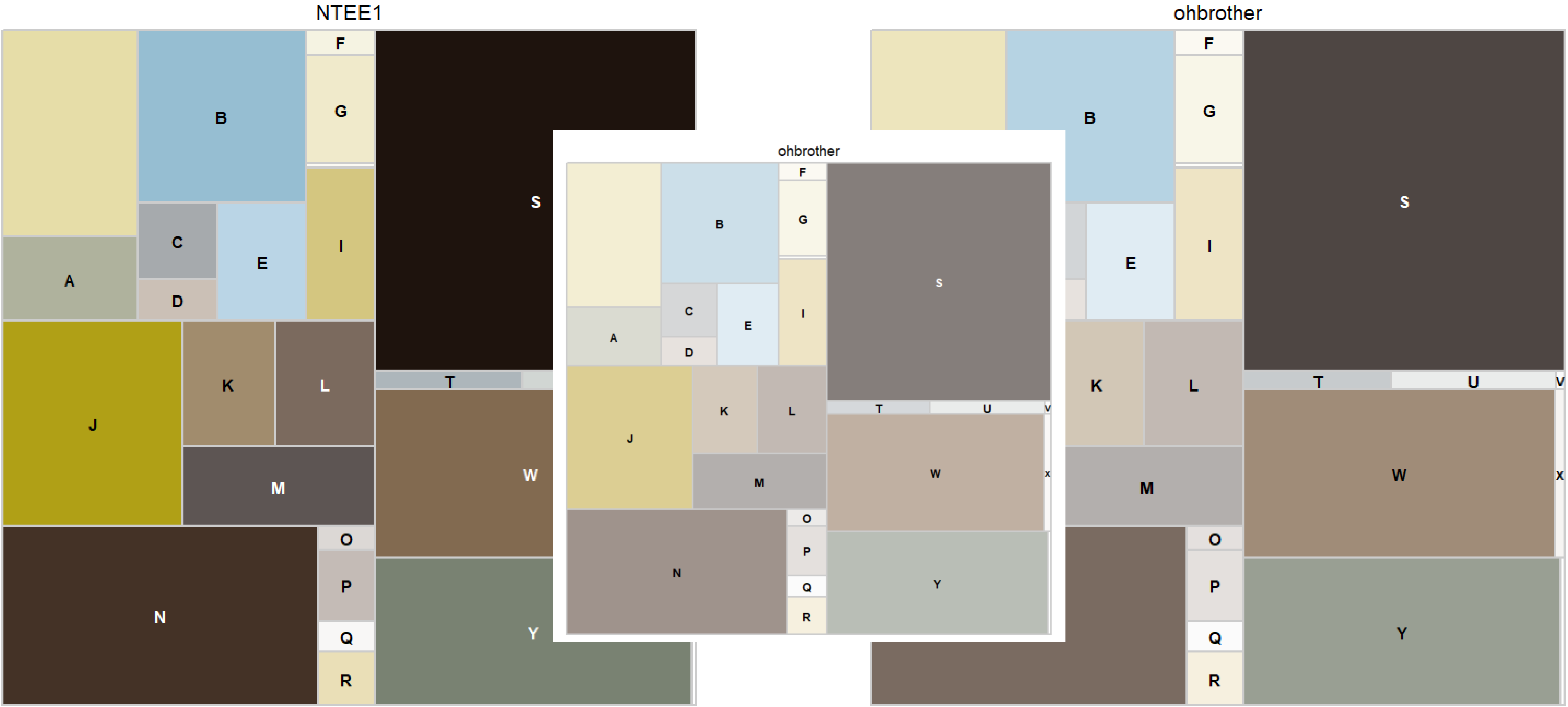
Formatting Experiments


Here are all of the pirate palettes
Transparency is set to 0
basel
pony
xmen
decision
southpark
google
eternal
evildead
usualsuspects
ohbrother
appletv
brave
bugs
cars
nemo
rat
up
espresso
ipod
info
info2

ipod



ohbrother



NTEE1
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ohbrother
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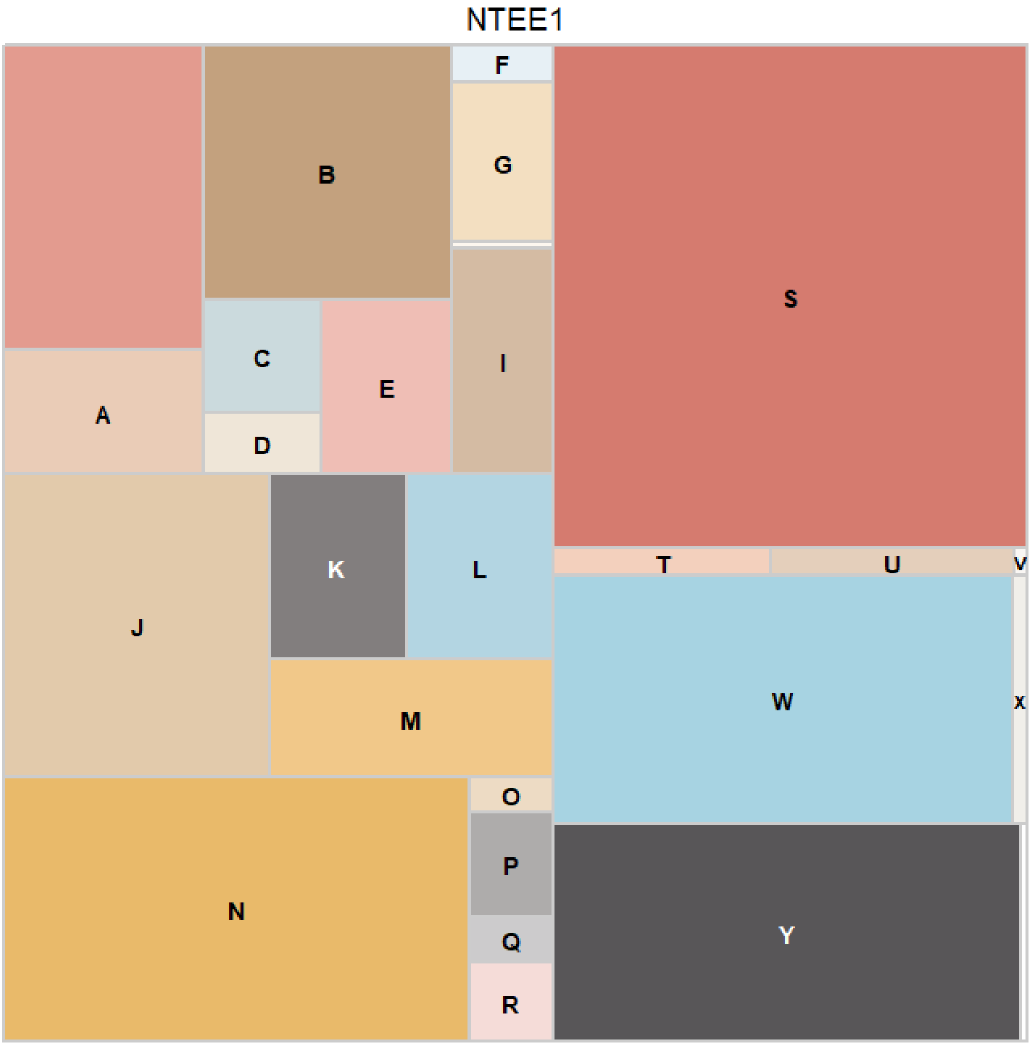
decision



NTEE1
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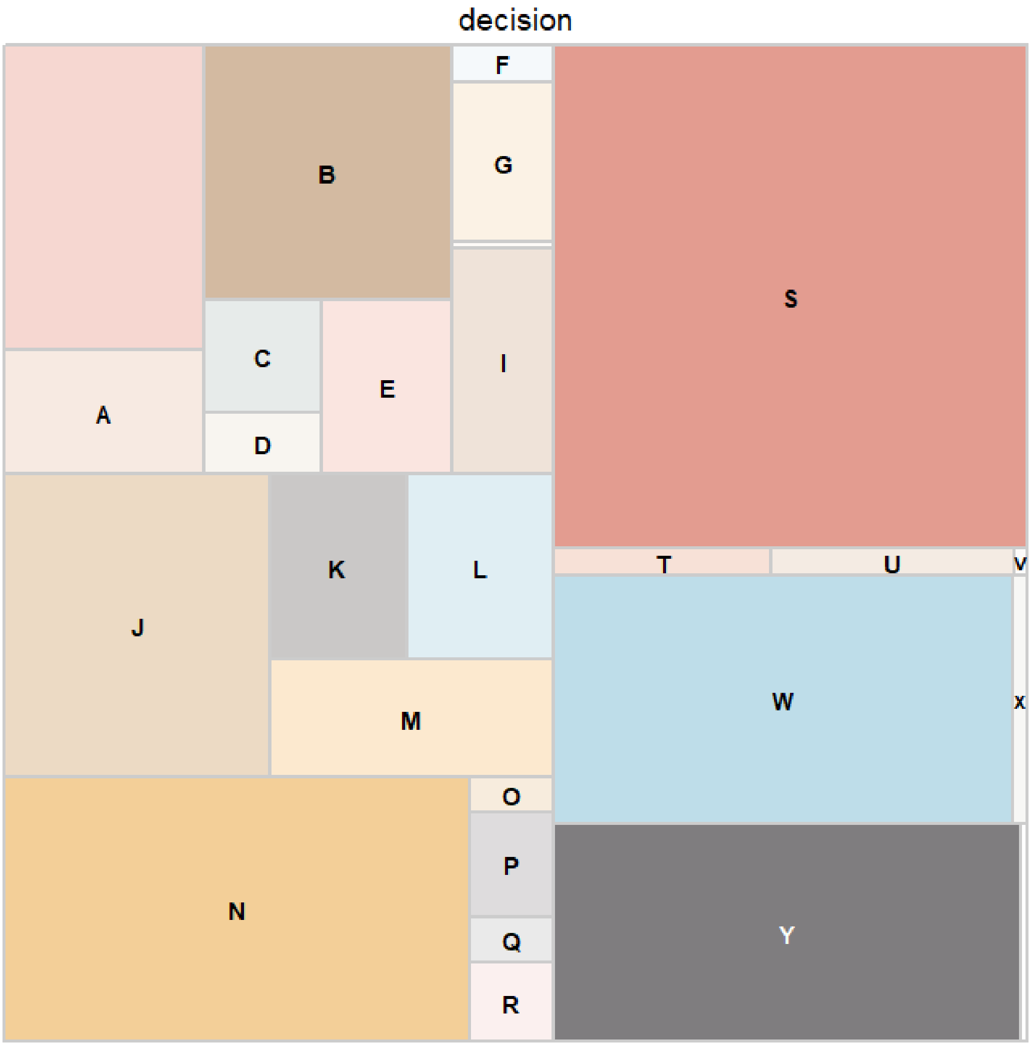

decision
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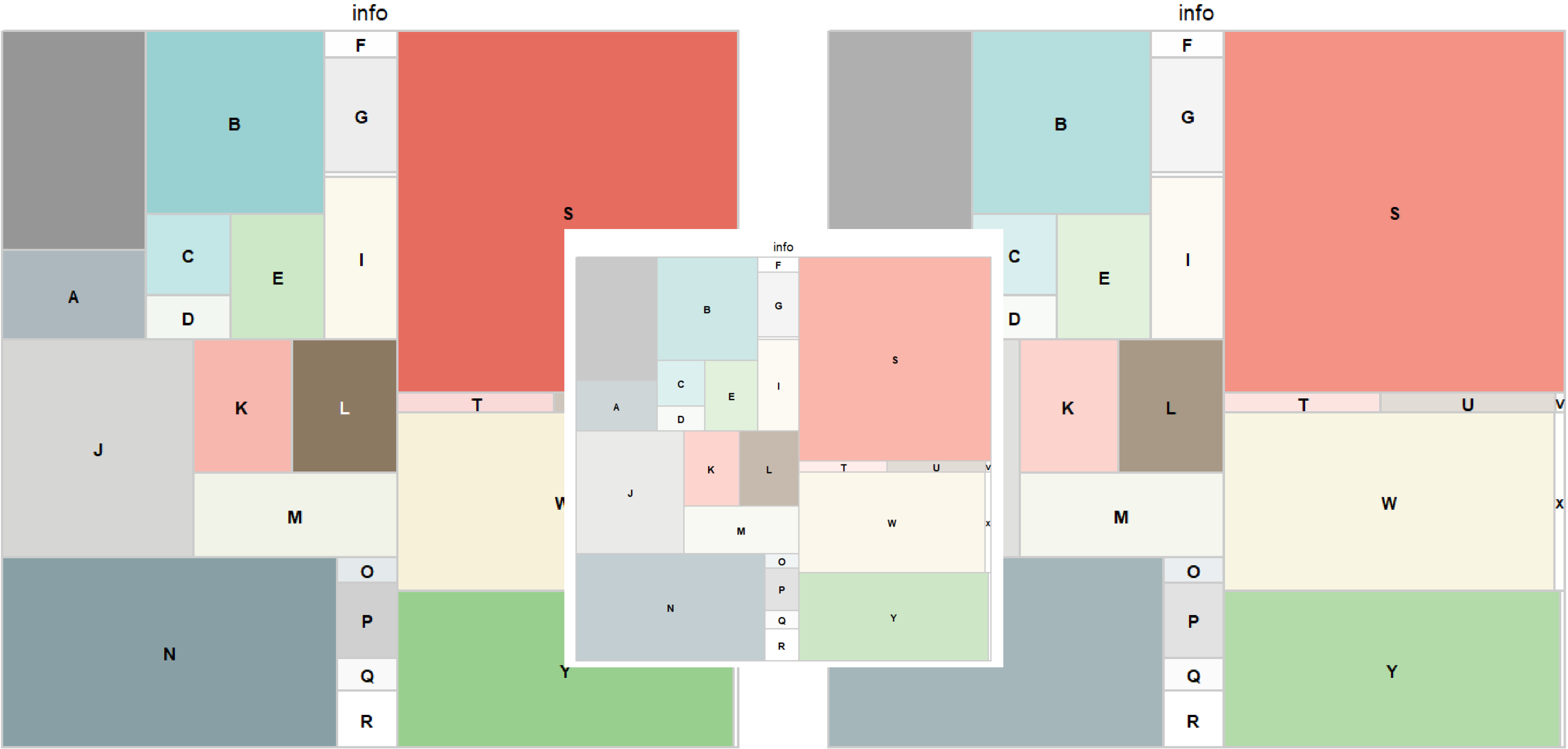
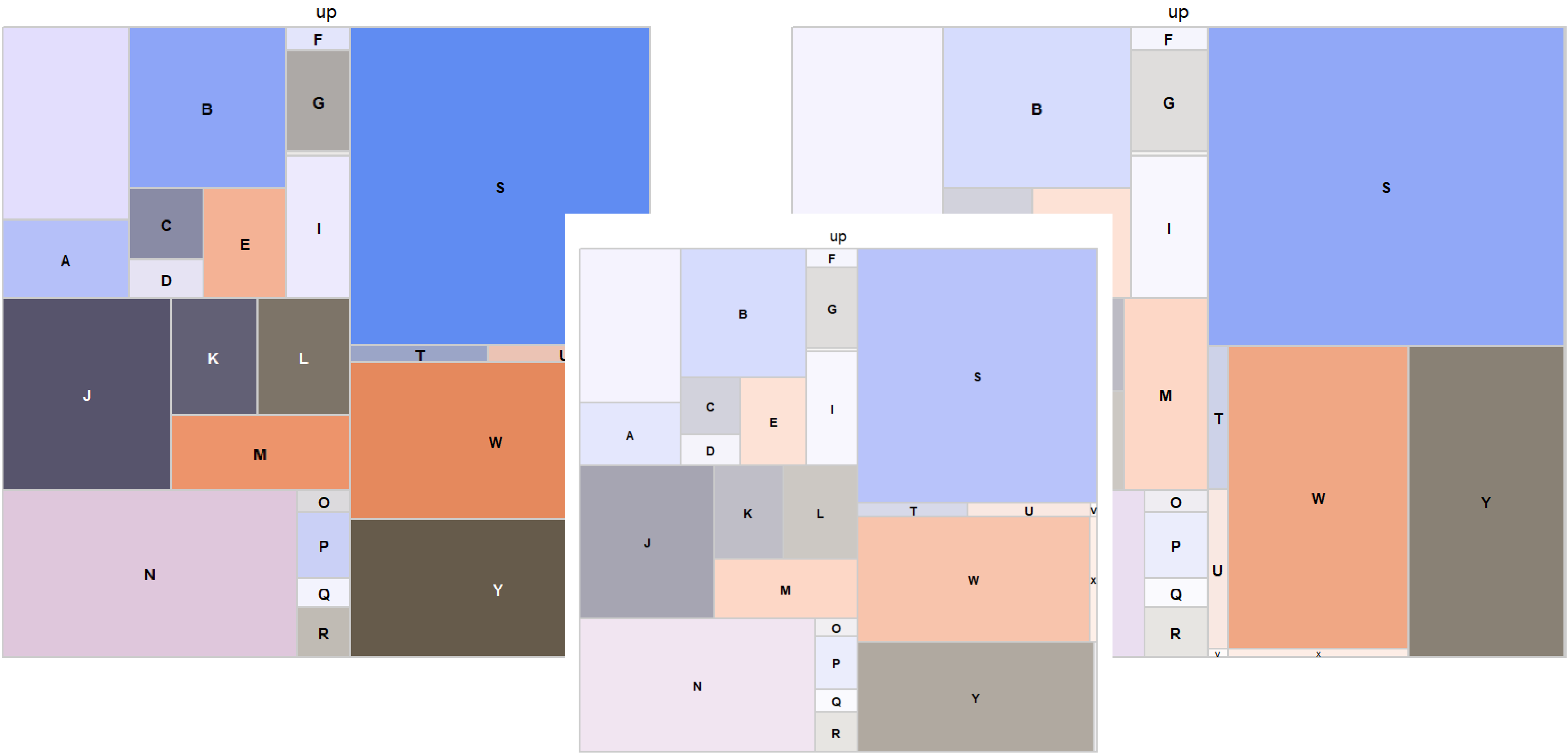
info




up
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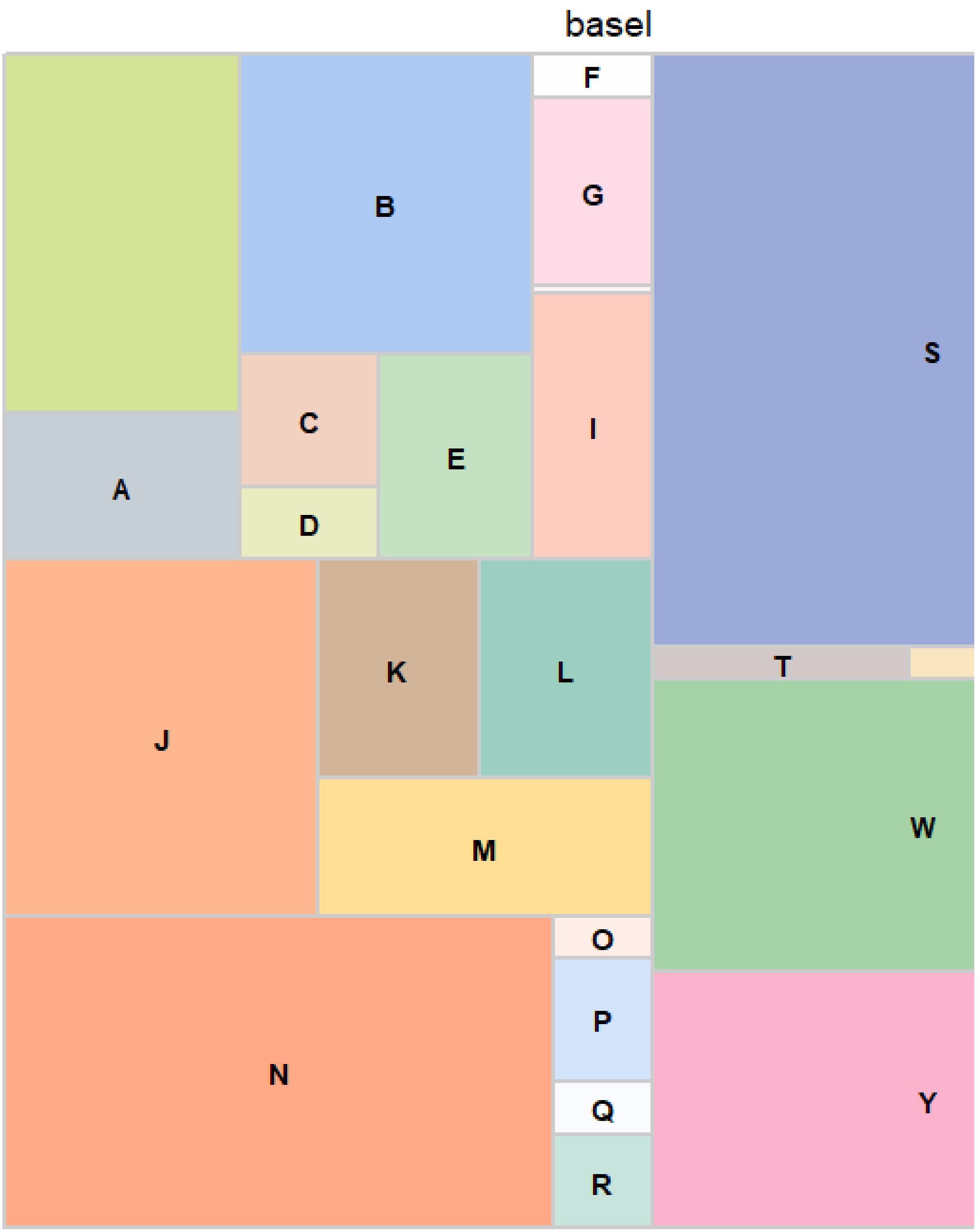
basel
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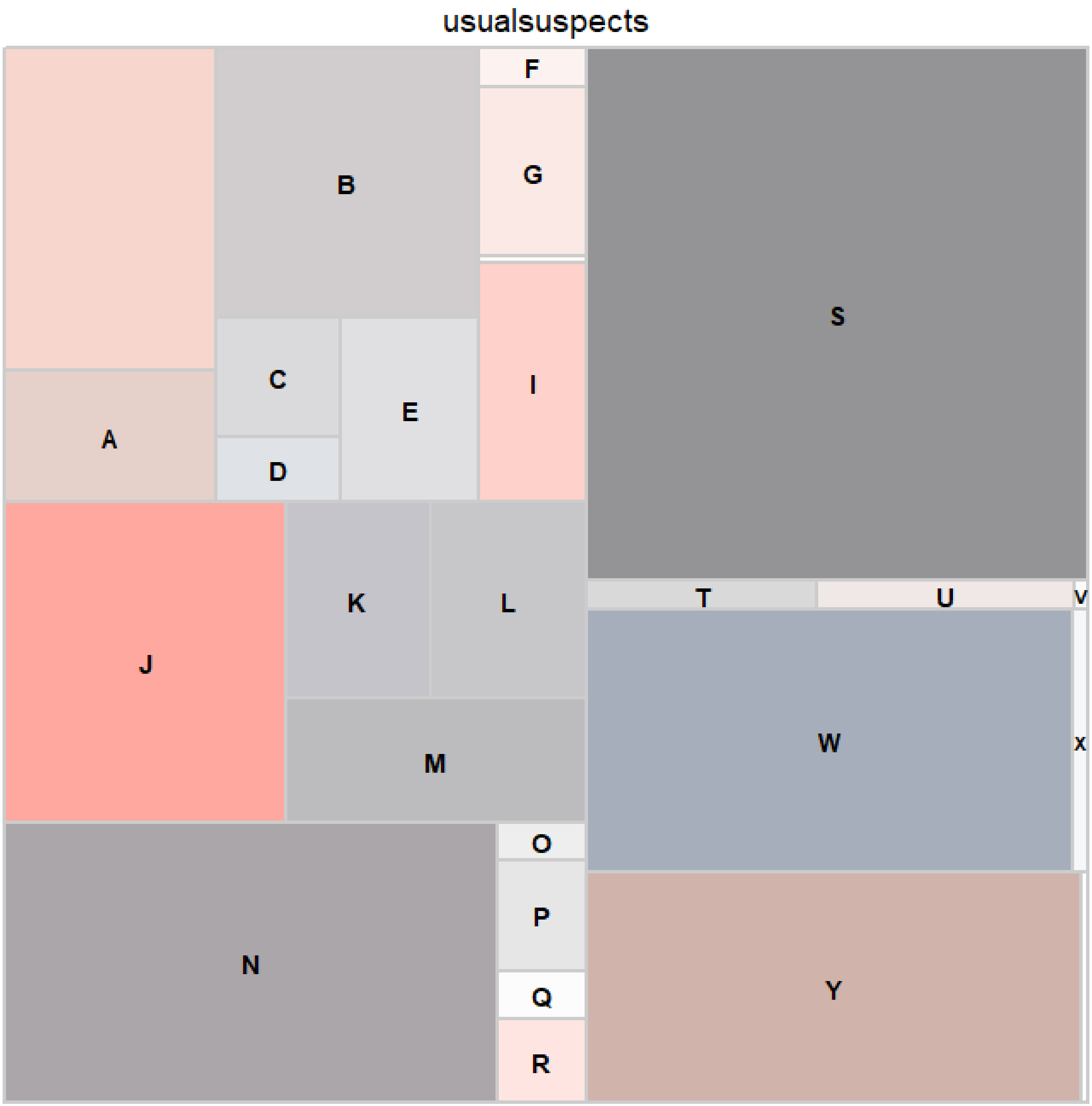
usualsuspects
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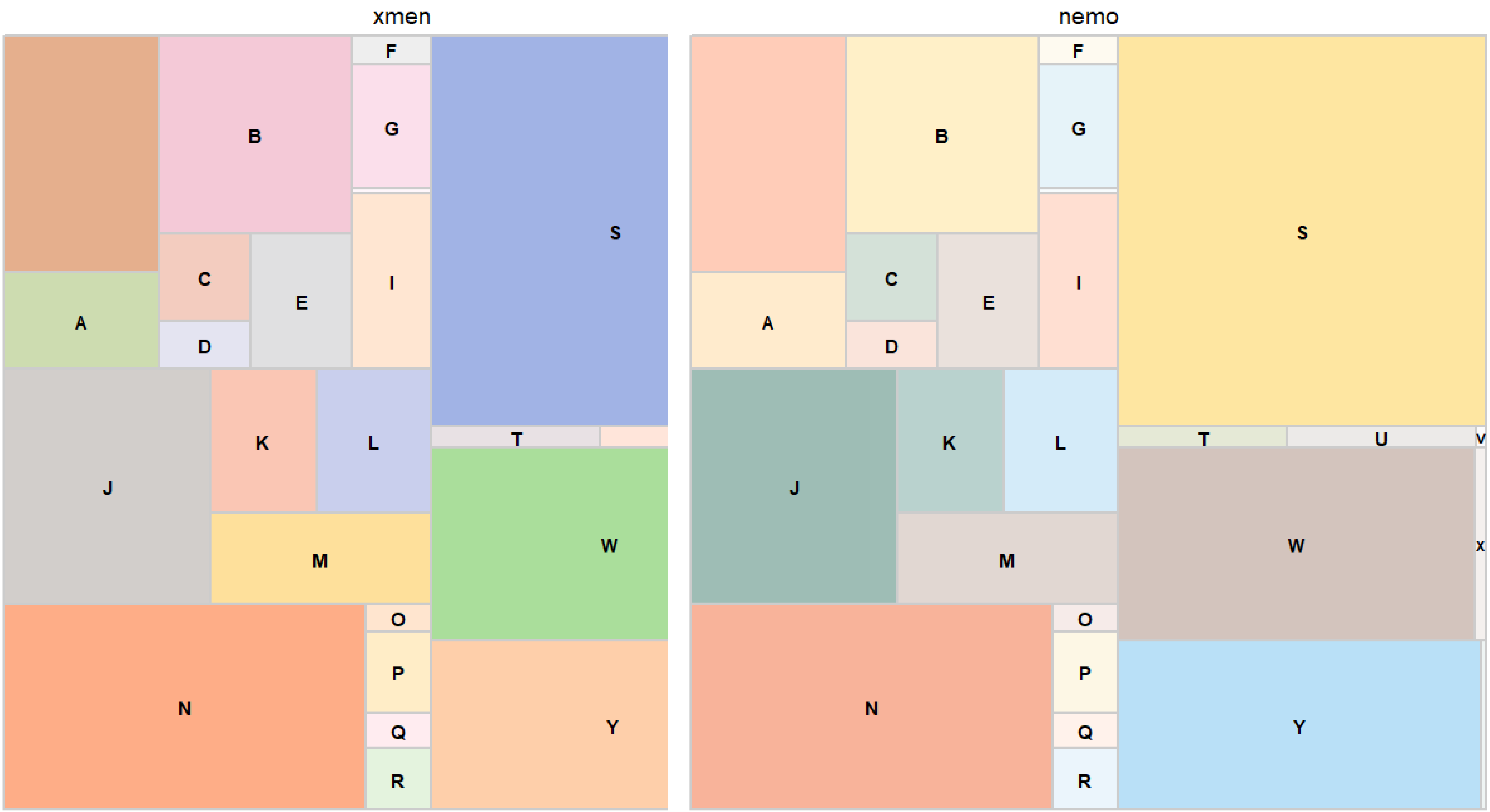
xmen
nemo
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